

TrES-3b

R-2736

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_190630 · created 2026-09-02T00:49:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458664.8262 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.005 +/- 0.056
Transit depth (Rp/Rs)^2	0.0026 +/- 0.058 [%]
Orbital Inclination (inc)	81.1 +/- 2.88
Airmass coefficient 1 (a1)	1.28 +/- 0.37
Airmass coefficient 2 (a2)	-0.22 +/- 0.92
Scatter in the residuals of the lightcurve fit is	28.69 %
Best Comparison Star	None
Optimal Aperture	2.96
Optimal Annulus	9.7
Transit Duration (day)	0.028 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.005 ± 0.056 vs 0.16309 (deviation 2.3σ)
Duration fit vs published	0.028 d vs 0.0567 d (deviation 0.9σ)
Mid-time O–C	2.5 min
Depth SNR	0.1
Residual scatter	28.69 %

Light curve

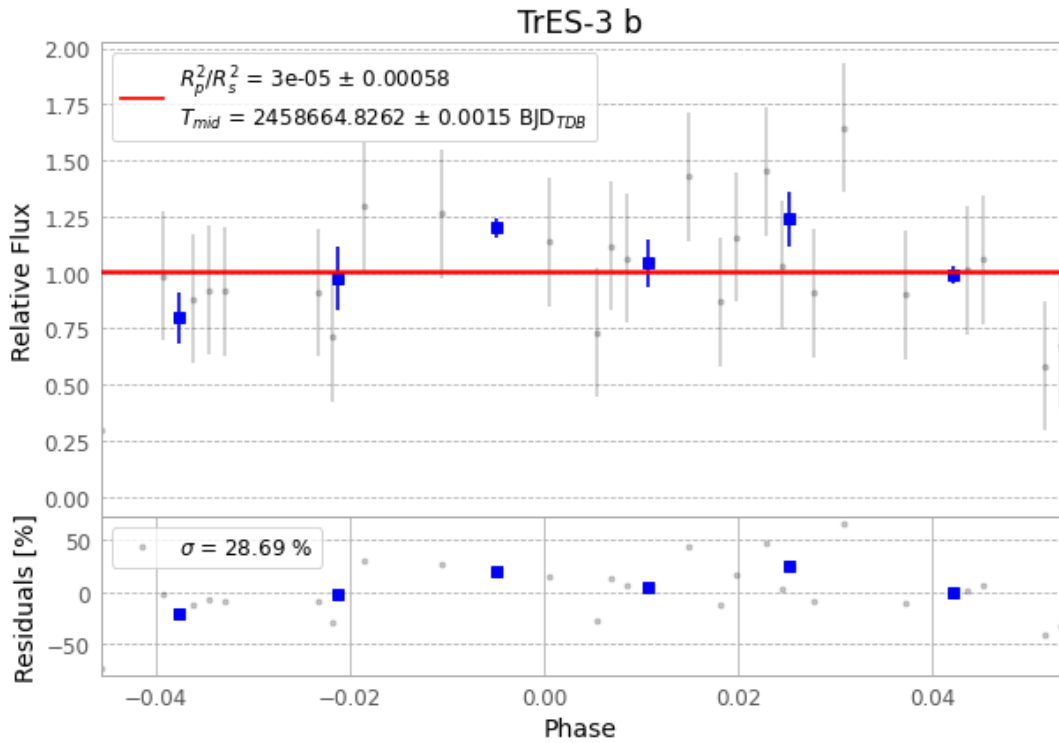


Figure 1 — FinalLightCurve_TrES-3 b_2019-06-29.png (SHA-256 181df63cf48e409a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [545, 206], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2f2b8071ecc5bddc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), TRES-3190630060615.FITS → TRES-3190630092412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-190630.log (1be7e5d64f19...), FinalLightCurve_TrES-3 b_2019-06-29.png (181df63cf48e...), FinalLightCurve_TrES-3 b_2019-06-29.csv (8fbcd415f3c1...)

Compute resources

Wall time 156.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 00:49Z.